

In this supplementary file we include the following information:

Figures S1 to S23

Tables S1 to S7

Section 1

1 INTRODUCTION

We include supplementary figures of the study area surface elevation, sample Sentinel 1 VV and VH time series, glacier hypsometry, crossover analysis for the airborne radar data for each year and compared to the surface-coupled ground penetrating radar in 2023, SWE values per elevation bin when including interpolation to Mt. Logan cores, SWE values per elevation bin without interpolation, Sentinel 1 images of the study region, two images of a snow pit wall, a plot of the snow density against depth, weather station time series plots of temperature and precipitation, CARRA2 temperatures, Sentinel 1 VV timeseries near the 2018 Kaskawulsh Core, and the radar amplitude for the last summer surface from each survey.

Data tables provide information regarding the individual snow pits and cores data as well as individual hyperbolic analysis surveys. We also include tables containing the values from plots elsewhere in this work regarding snowline/ELA elevation and accumulation area mass input.

Lastly, we include a section that discusses the propagation of uncertainty for calculations of snow depth and real permittivity from two way travel time, the uncertainty for the 50 m elevation bins, and the uncertainty for the accumulation area mass input.

19 2 SUPPLEMENTARY FIGURES

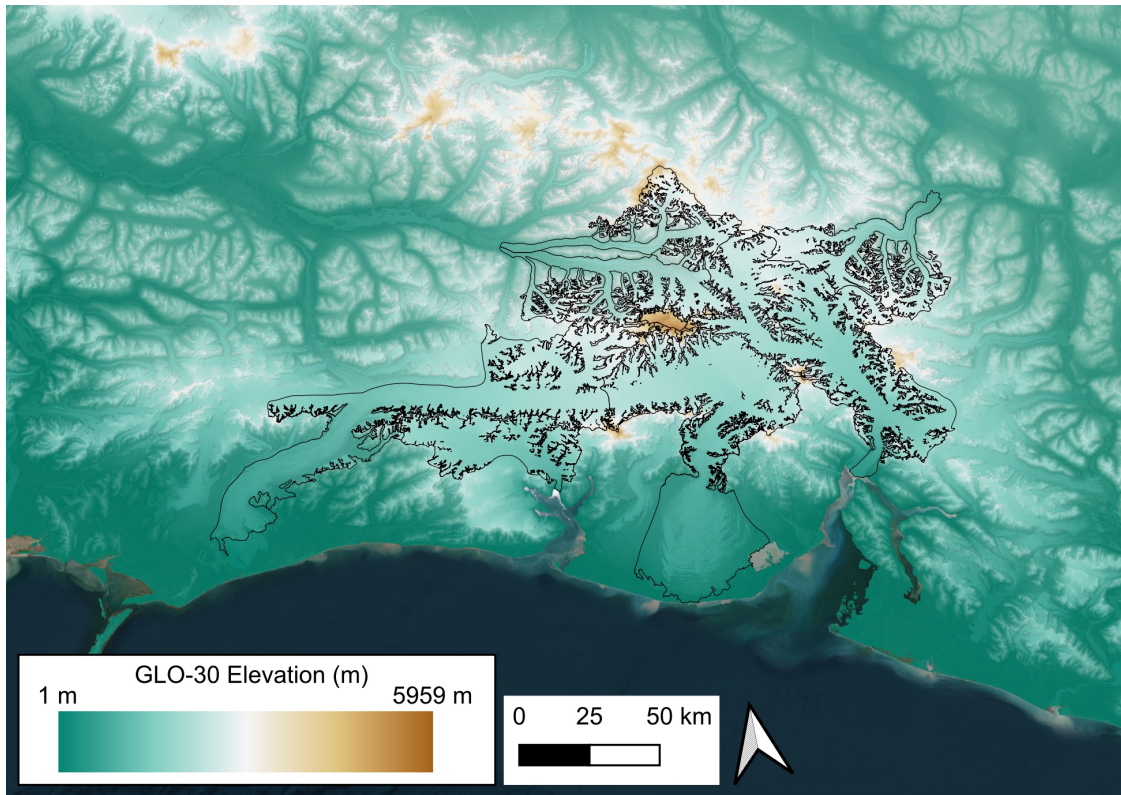


Fig. S1. Surface elevation of the St. Elias Mountains from the Copernicus GLO-30 digital elevation model. The glaciers in our study are outlined in black. Elevations reach a maximum of 5959 m a.s.l. at the summit of Mt. Logan. Data from OpenTopography (2021).

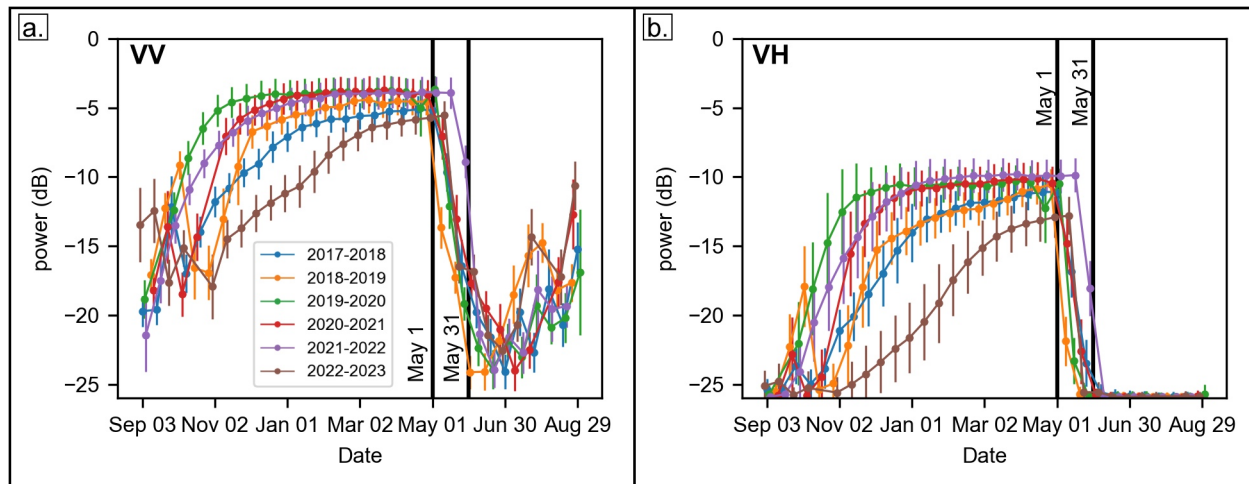


Fig. S2. Time series of Sentinel 1 (a) VV and (b) VH reflectivity at the 2023 Seward core site. Error bars are one standard deviation of the power in the sampling area. Similar features are visible for both polarizations, but with a lower returned power by ~ 6 dB for VH. While the features in May were observable in both polarizations, features in the early fall are less prominent in VH. Additionally, the VH polarization dips below the noise equivalent sigma zero (-26 dB) for much of the summer, whereas the VV polarization does not.

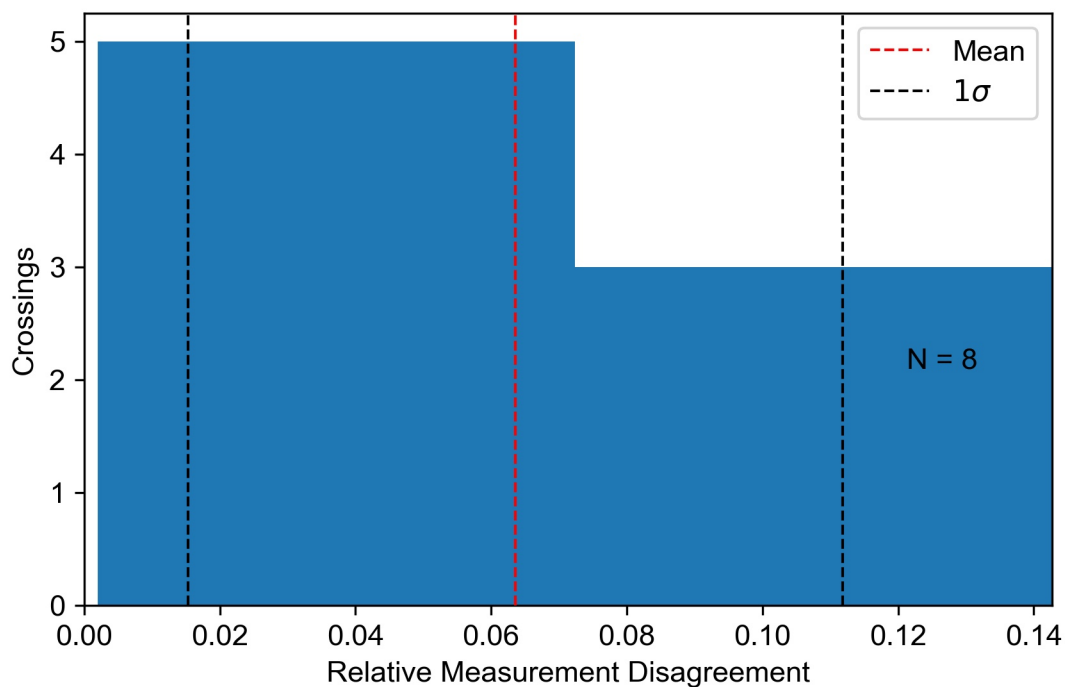


Fig. S3. Spread of the 2018 crossover points in the relative measurement space.

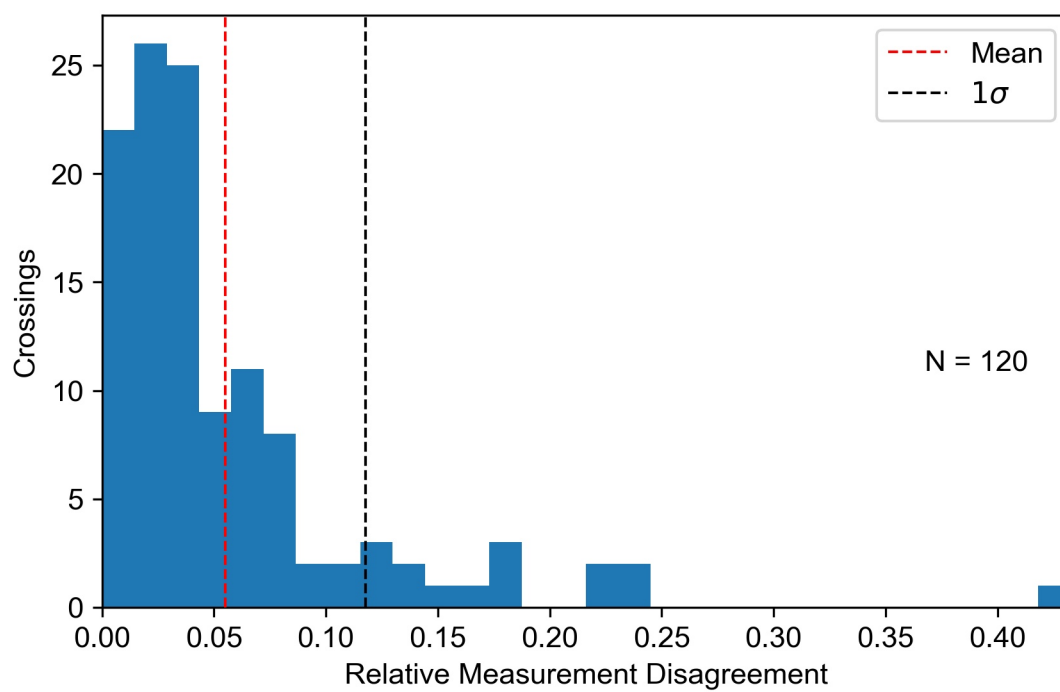


Fig. S4. Spread of the 2021 crossover points in the relative measurement space. Two thirds of the 2023 crossover points fall below the mean.

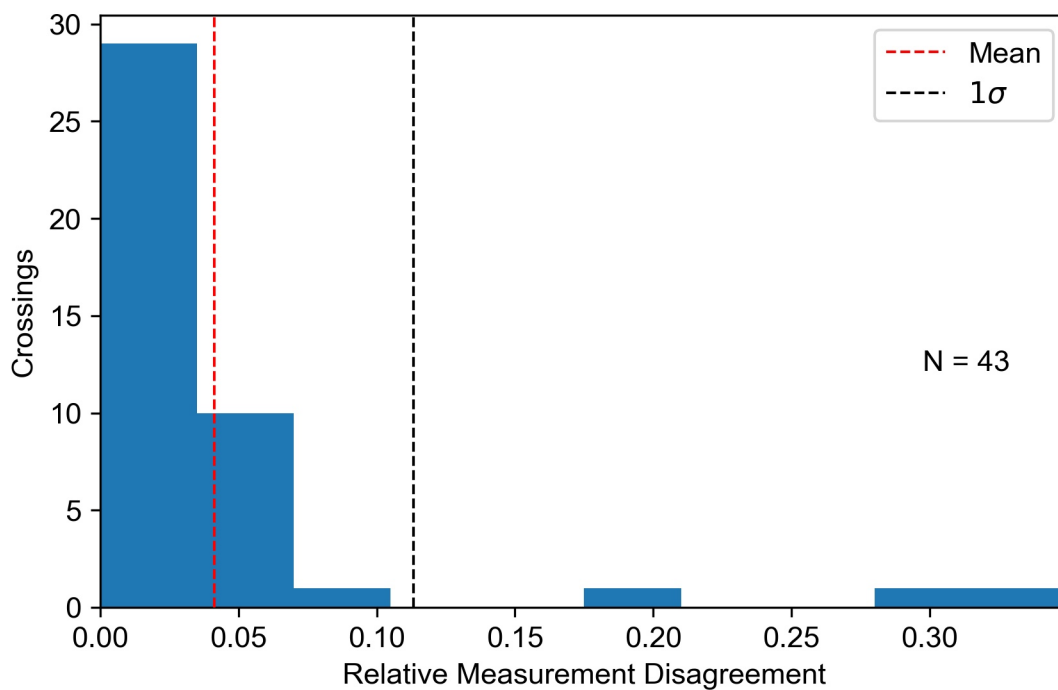


Fig. S5. Spread of the 2023 crossover points in the relative measurement space. Two thirds of the 2023 crossover points fall below the mean. The extremely different points tend to be in areas where tracking reflectors for long stretches was challenging or impossible. Thus, large differences are likely due to the picking the shallower reflector in radargram and the last summer surface in the other due to the similarities when both reflectors were not present.

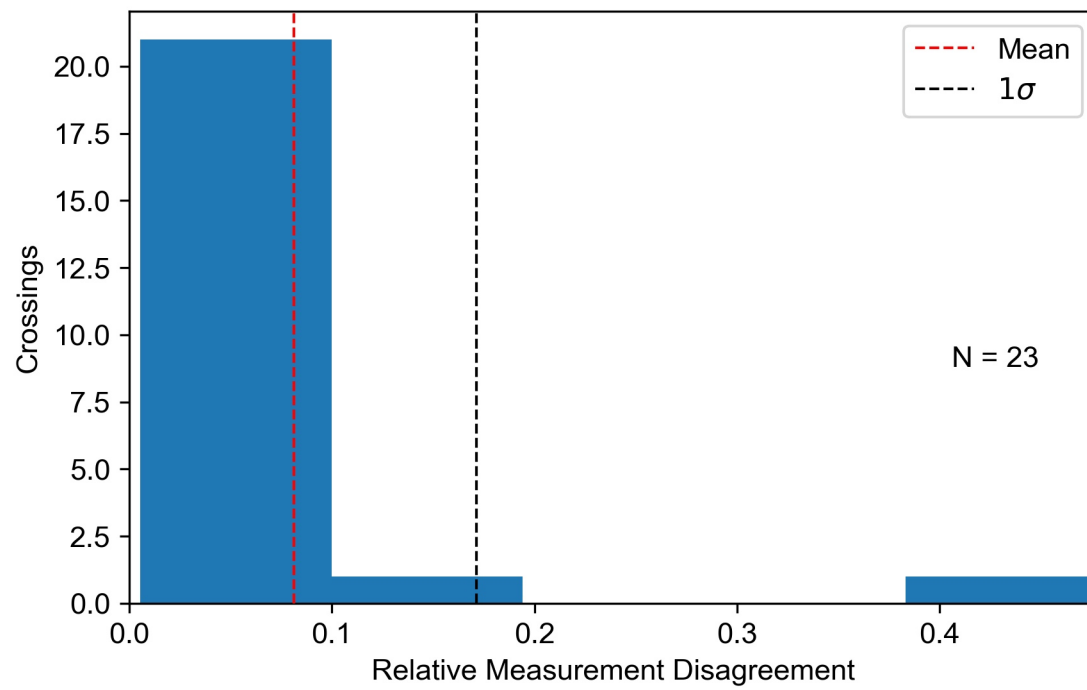


Fig. S6. Spread of the 2023 crossover points between the surface GPR and the airborne radar. At all but two crossover points, the airborne radar was deeper than the surface GPR.

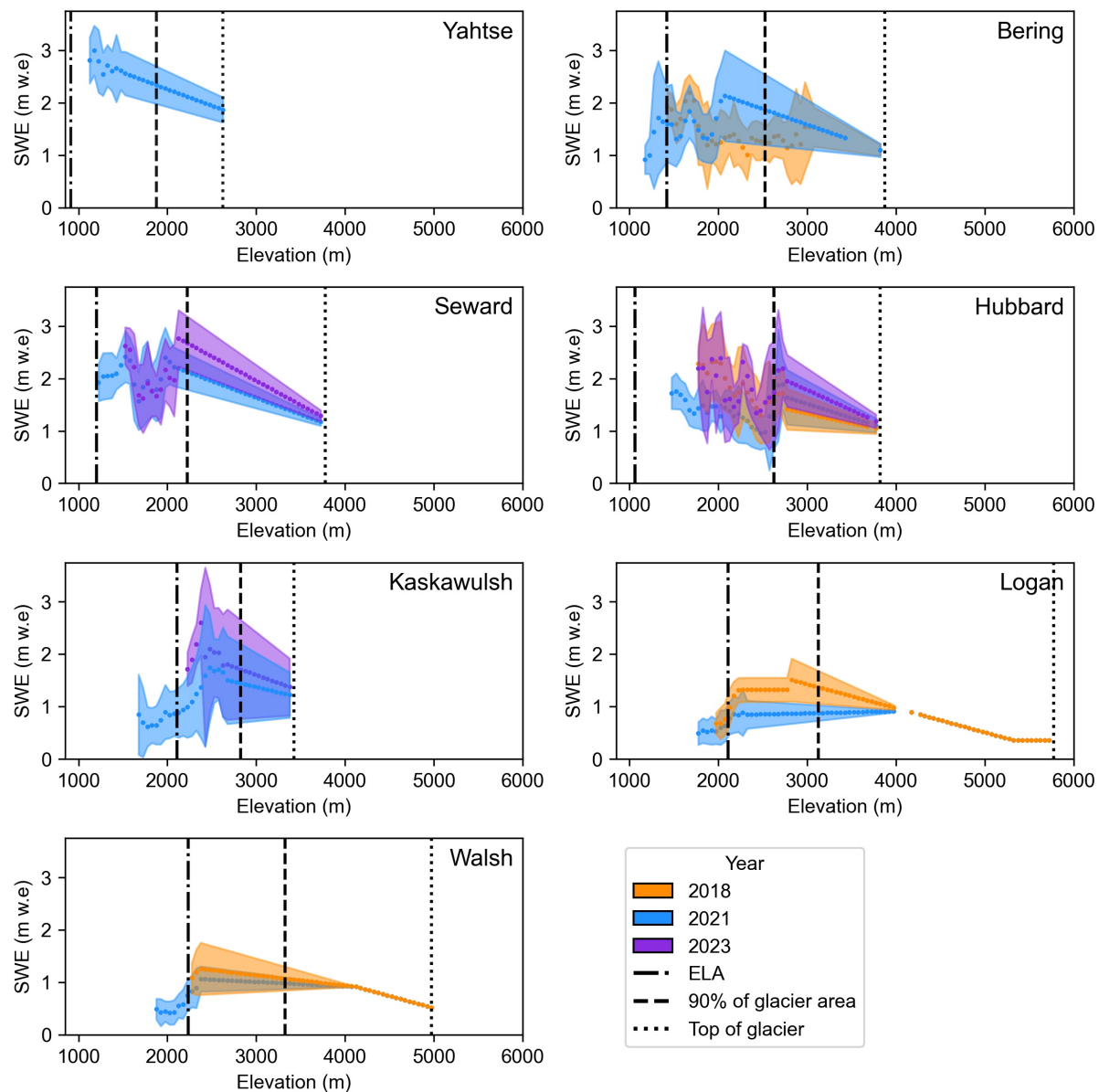


Fig. S7. SWE values for each elevation bin including the interpolation to the high elevation core sites on Mt. Logan.

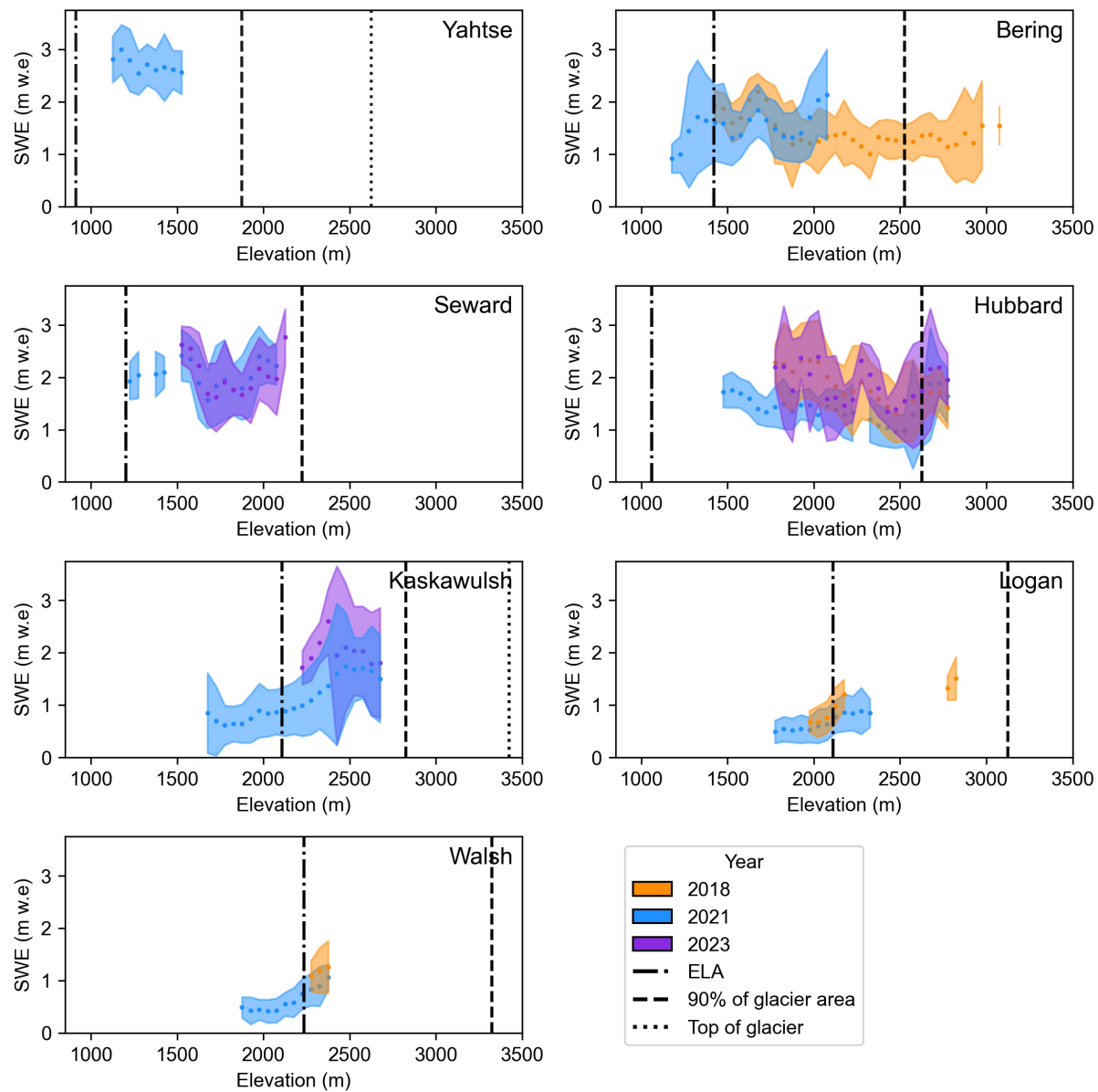


Fig. S8. SWE values for each elevation bin without the interpolation of missing data within elevations sampled by the airborne radar. Interpolation within the radar data only needed to be performed of the Seward in 2021, Hubbard in 2021, and Logan in 2018.

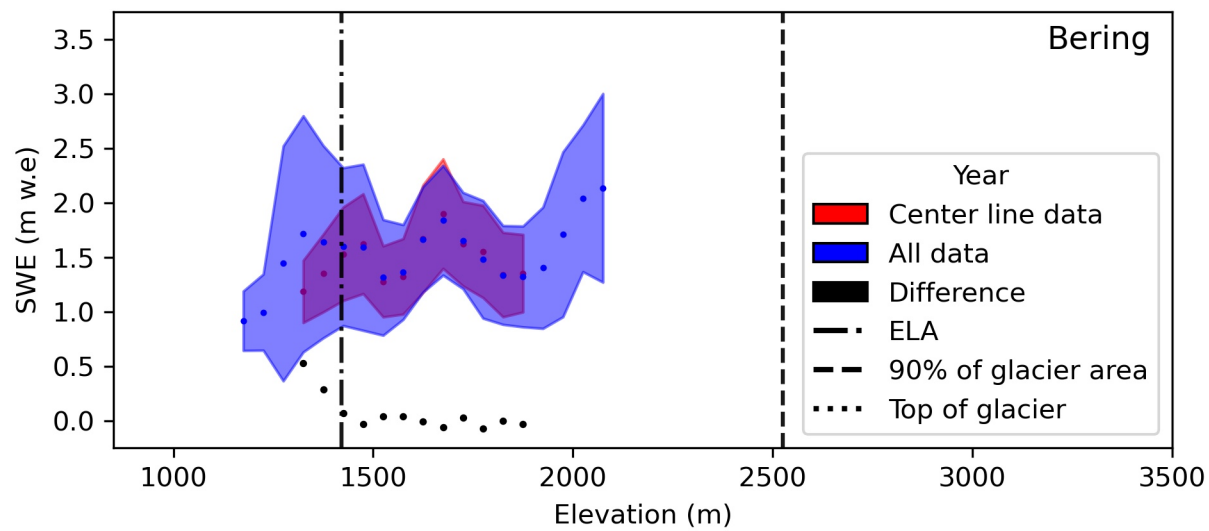


Fig. S9. SWE values for the 2021 radar data within 1 km of the Bering Glacier center line plotted against all the Bering Glacier data, and the difference between the two as black points. For most of the elevation bins, the difference between the center line data and all the data is near zero, and both points fall within the uncertainty of the other point. The difference is calculated as the all data SWE minus the center line data SWE per elevation bin.

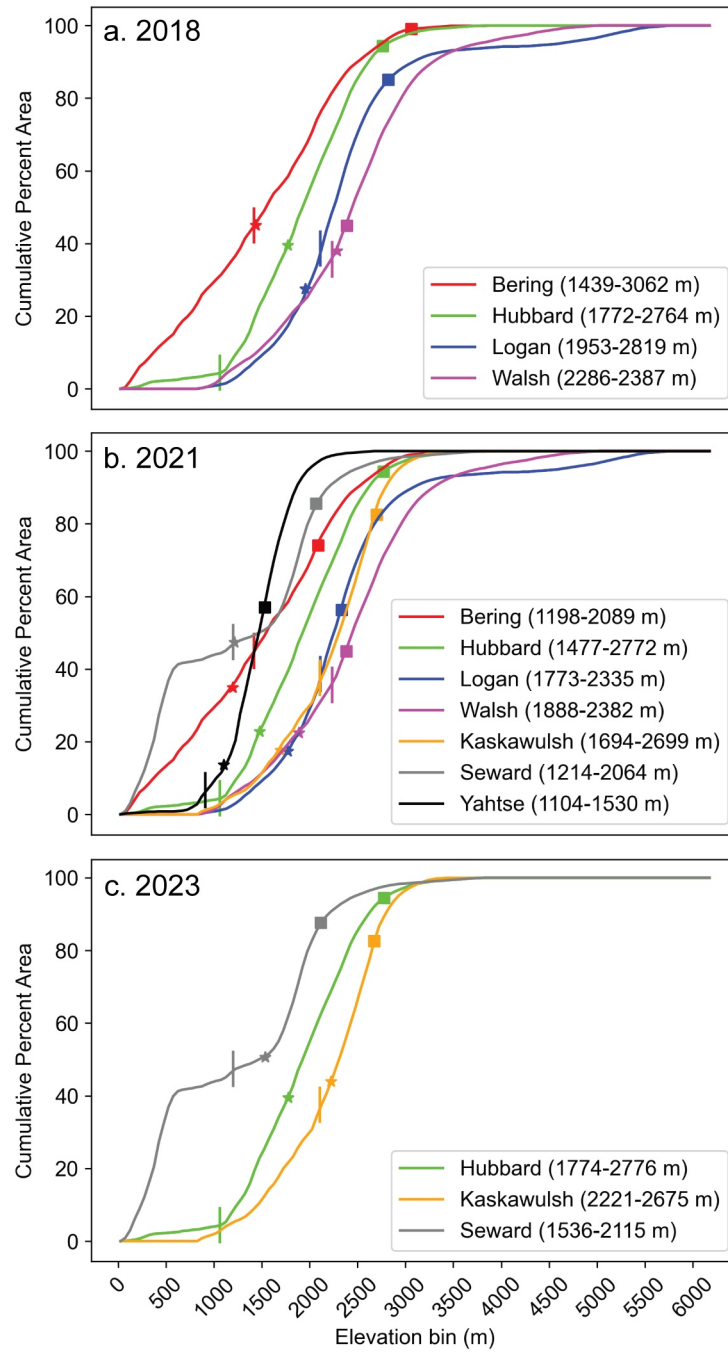


Fig. S10. Cumulative percent of the glacier area at each elevation bin for (a) 2018, (b) 2021, (c) 2023. Each glacier is marked by the lowest (star) and highest (square) surface elevation with airborne data and the mean ELA (vertical line). Glacier labels in the legend are the surface elevation range for the airborne data. Relevant numerical values are listed in Table S4.

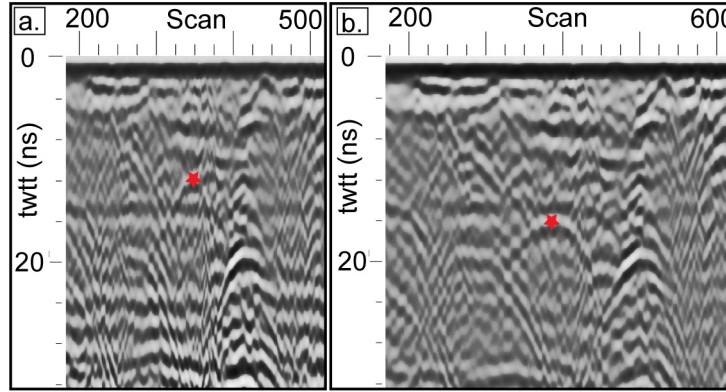


Fig. S11. Hyperbolic analysis surveys for the probe at depths of (a) 1 m and (b) 1.5 m. The vertex of each hyperbola caused by the probe is marked by a red star. The column of parabolas at roughly scan 400 in (a) and 500 in (b) did not change as the probe depth was changed. Thus we attribute these parabolas to the snow pit wall.

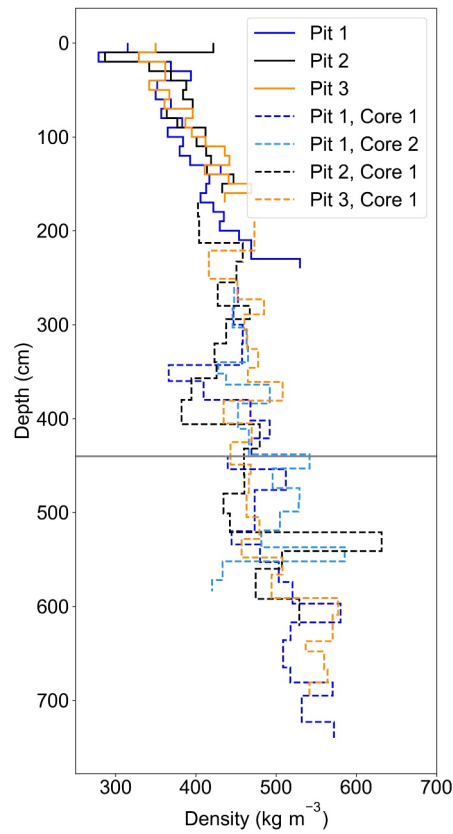


Fig. S12. Measured densities for the May 2023 Seward pits (solid lines) and cores (dashed). Locations of each pit are reported in Table S1. We omit the 9% uncertainty on each measurement in the plot for ease of seeing the measured value. Discontinuities at the pit to core interface are attributed to the change in sampling techniques. The gray horizontal line marks the approximate last summer surface depth.

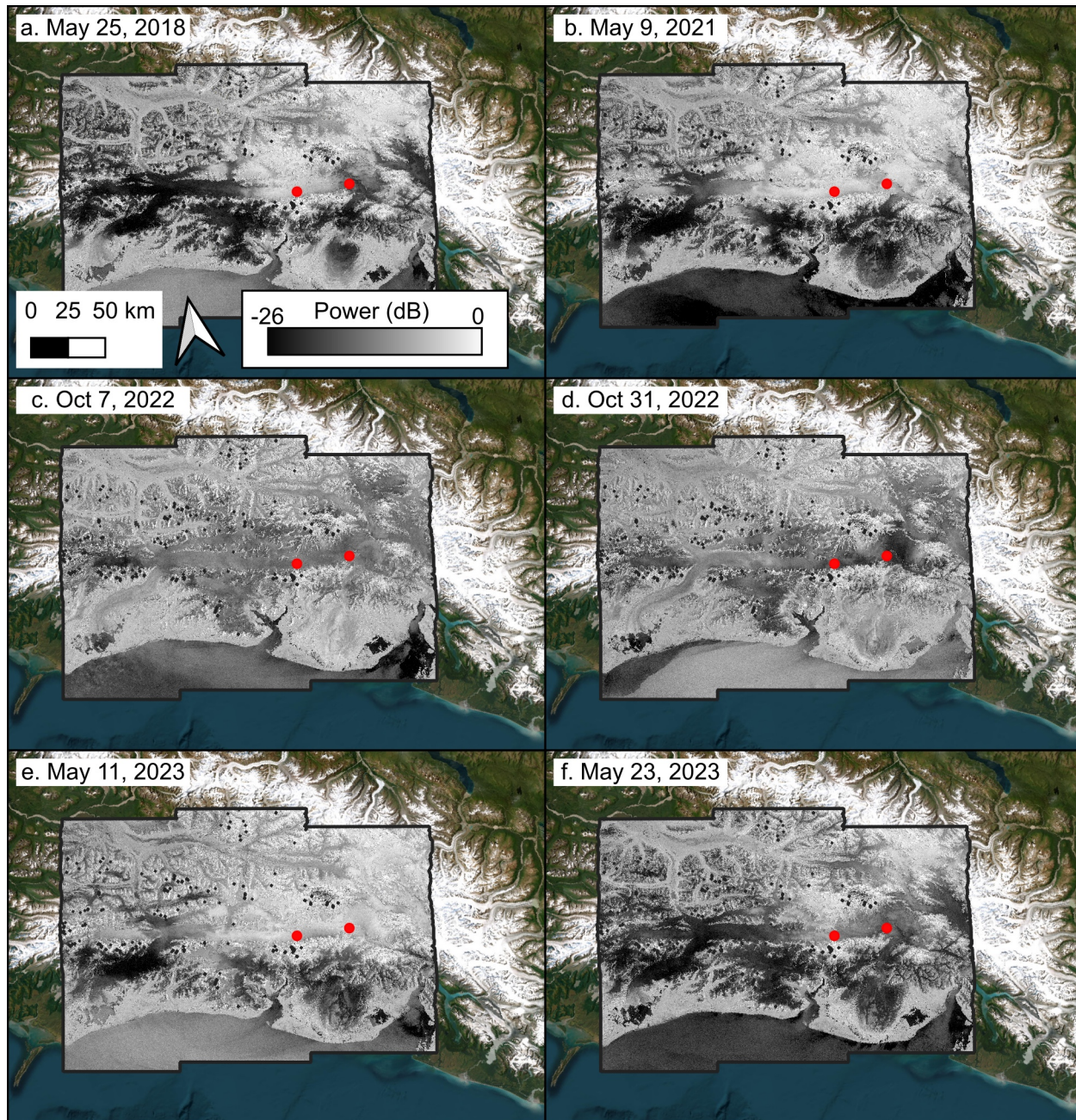


Fig. S13. VV polarized Sentinel 1 images from the Alaska Satellite Facility approximately during airborne radar acquisition in (a) 2018 and (b) 2021, (c, d) during early Fall 2022, (e) during 2023 surface GPR acquisition, and (f) during the 2023 airborne radar acquisition. The spatial scale and color bar is the same between all panels so we only include them in panel (a). Darker colors represent lower reflectivity. Color scale is cropped to -26 dB to 0 dB to match the scale in 10. The two points marked on all panels are the sampling locations in Fig. 10 at the 2023 Seward core site and the Seward-Bering ice divide. The black outline is included to mark the edge of the Sentinel 1 image. ASF DAAC 2015, contains modified Copernicus Sentinel data 2015, processed by ESA.

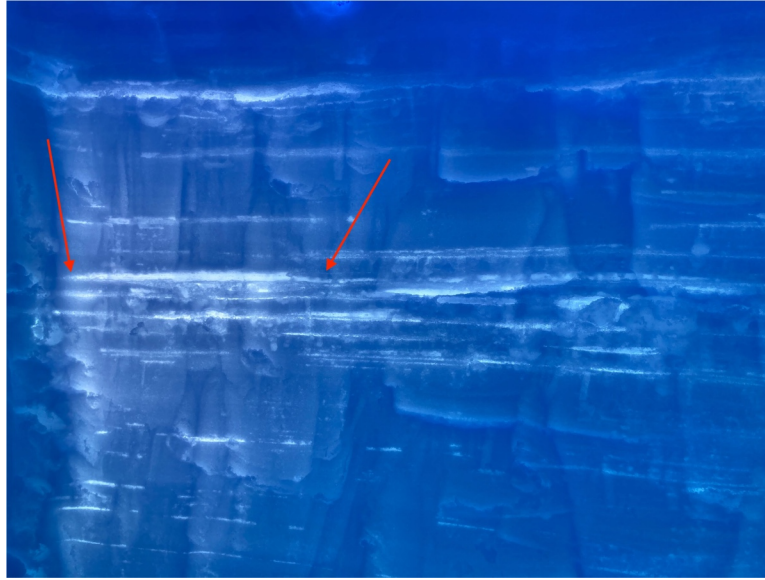


Fig. S14. Upper section of the May 2023 Seward Snow Pit 1 with the location reported in Table S1. Photo was taken by digging two adjacent snow pits with a preserved snow wall between them. The blue color is due to a blue tarp covering the snow pit. The ice layers were much more apparent through the snow wall than looking at the wall of a single snow pit. Red arrows in Figs. S14 and S15 are the same ice lens.

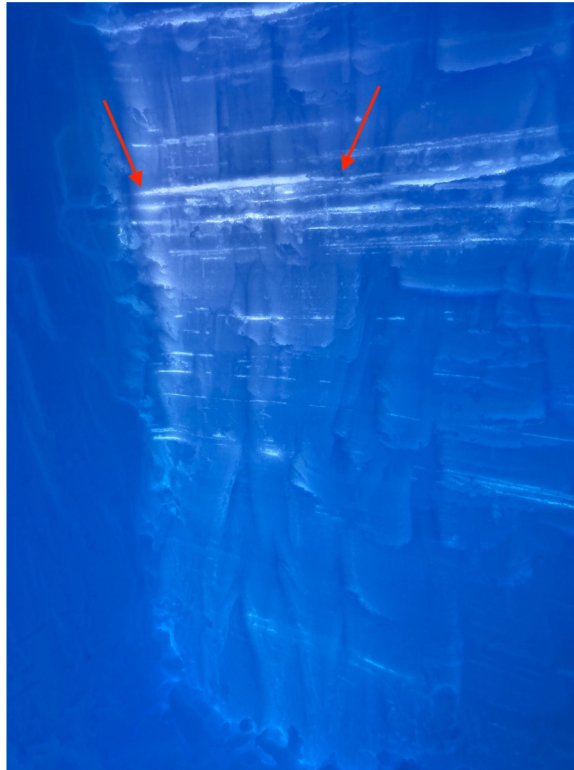


Fig. S15. Deeper section of the May 2023 Seward Snow Pit 1 with the location reported in Table S1. The bottom of the snow pit is in the lower left of the image. After the concentration of ice layers shown in Fig. S14, they become less common deeper in the snow column.

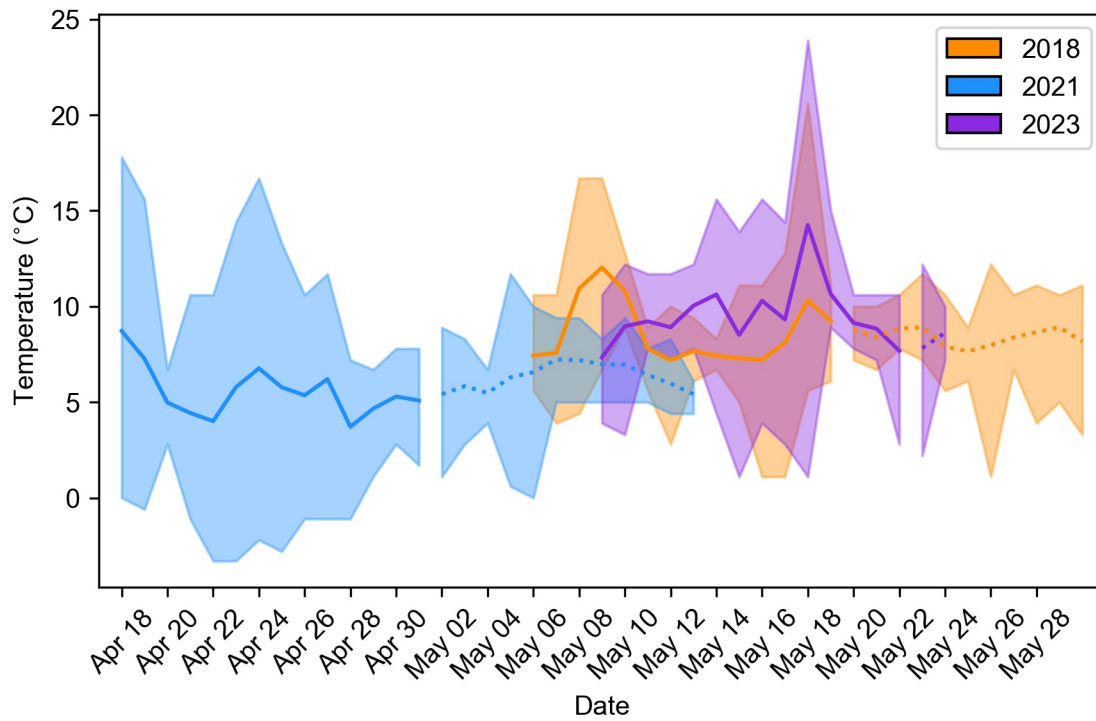


Fig. S16. Daily temperature time series for Yakutat Airport (10 m a.s.l.). Solid line is the daily mean, with the dotted lines being the daily mean during airborne data acquisition. Shading ranges between the daily minimum and maximum temperature. Yakutat is the most continental of the three weather stations.

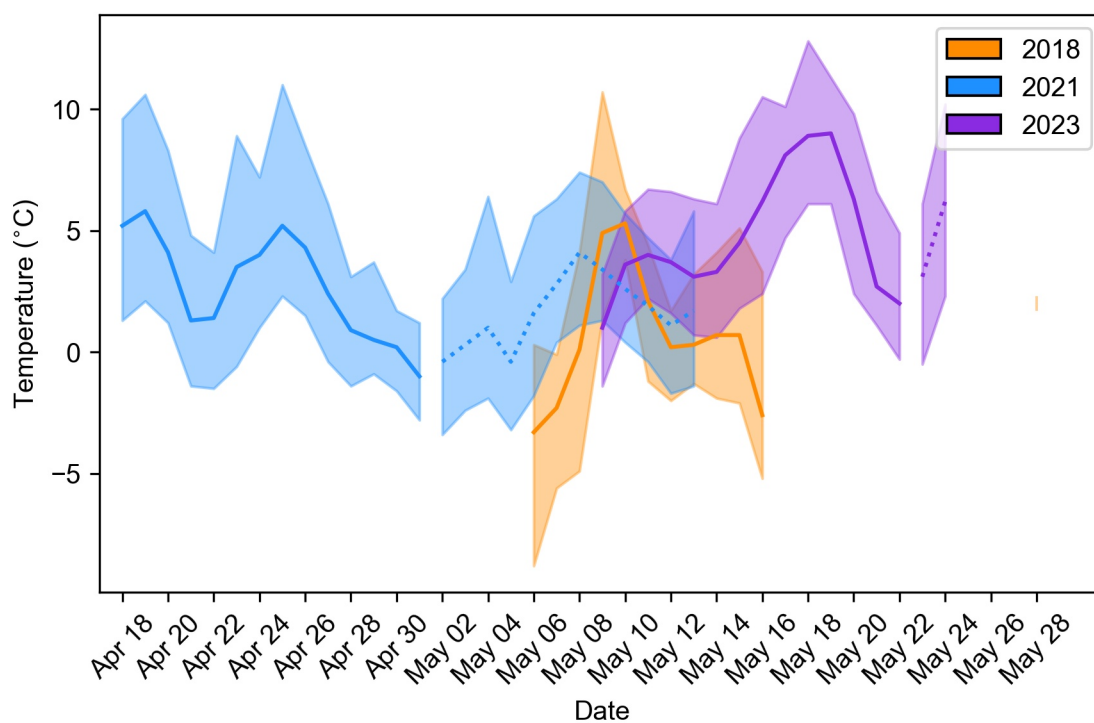


Fig. S17. Daily temperature time series for Tana Knob (1050 m a.s.l.). Solid line is the daily mean, with the dotted lines being the daily mean during airborne data acquisition. Shading ranges between the daily minimum and maximum temperature. Tana Knob was missing data in 2018, which is why there is less data coverage than compared to Yakutat (Fig. S16). Tana Knob is a maritime/continental weather station, it is inland of the first range of coastal mountains in the St. Elias range, but not as continental as Haines Junction.

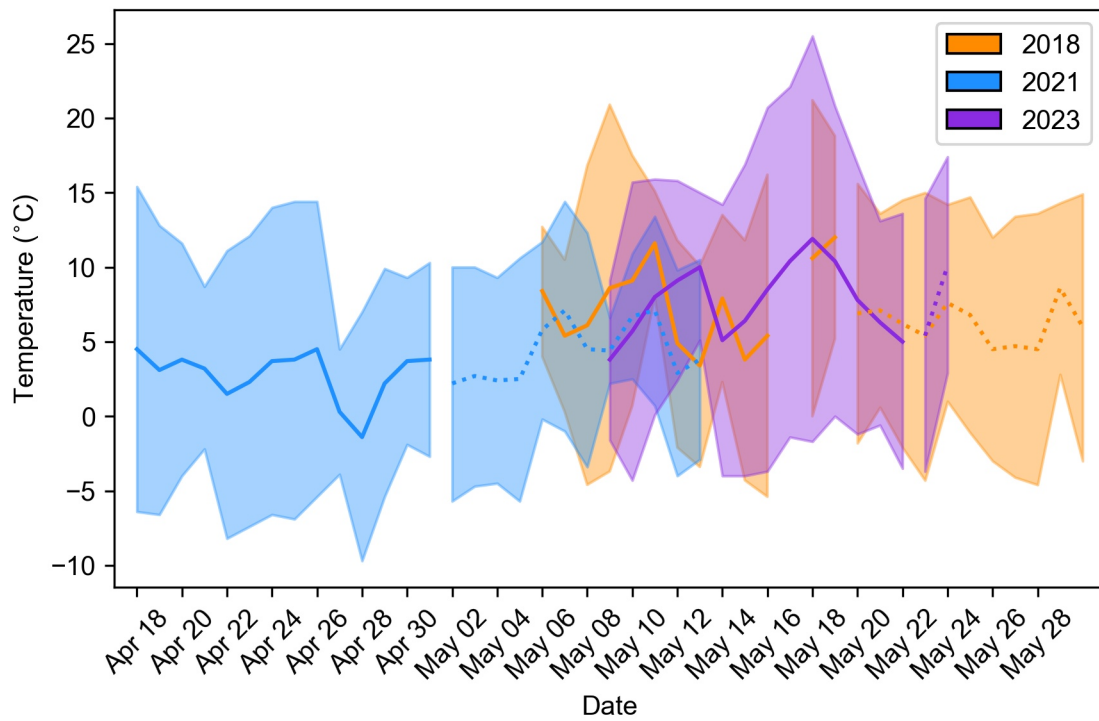


Fig. S18. Daily temperature time series for Haines Junction (595 m a.s.l.). Solid line is the daily mean, with the dotted lines being the daily mean during airborne data acquisition. Shading ranges between the daily minimum and maximum temperature. Tana Knob was missing data during May 2018 resulting in an incomplete data record. Haines Junction is the most continental of the three weather stations.

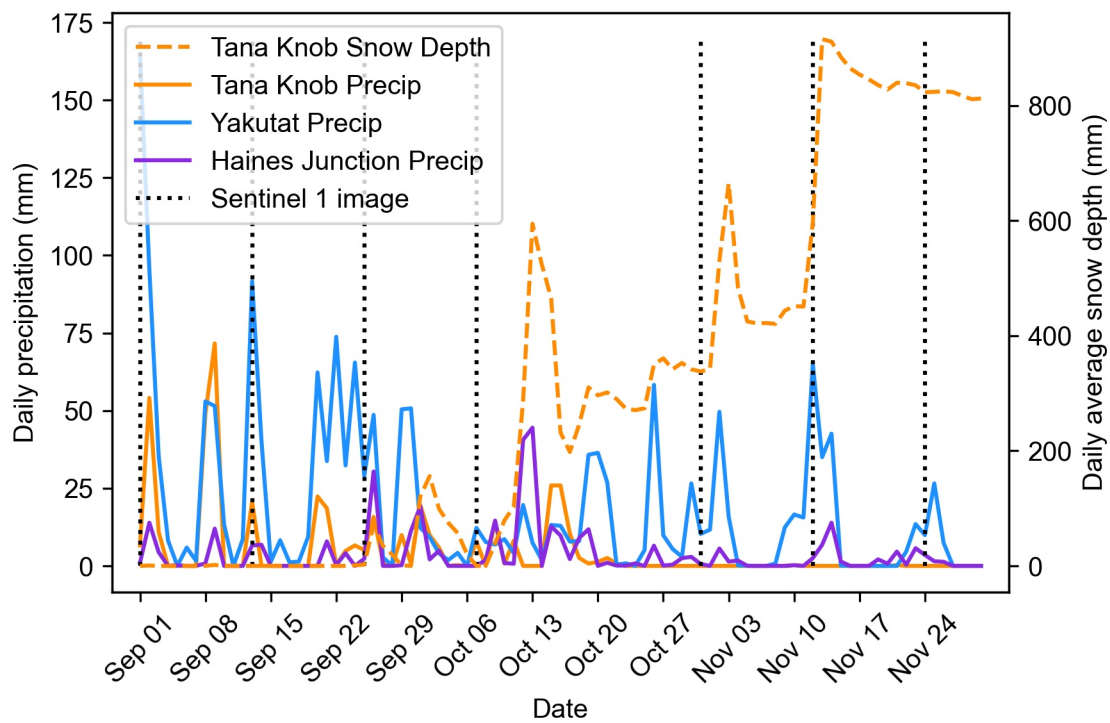


Fig. S19. Daily precipitation total (solid lines) for Tana Knob, Yakutat, and Haines Junction and the daily mean snow depth (dashed line) at Tana Knob for September 1st through November 30th, 2022. The lack of precipitation at Tana Knob after mid-October is likely due to no melt or liquid precipitation. The vertical dotted lines mark days when Sentinel 1 imagery was collected for the selected path and frame. There was no mid-October, 2022 image for the selected path and frame.

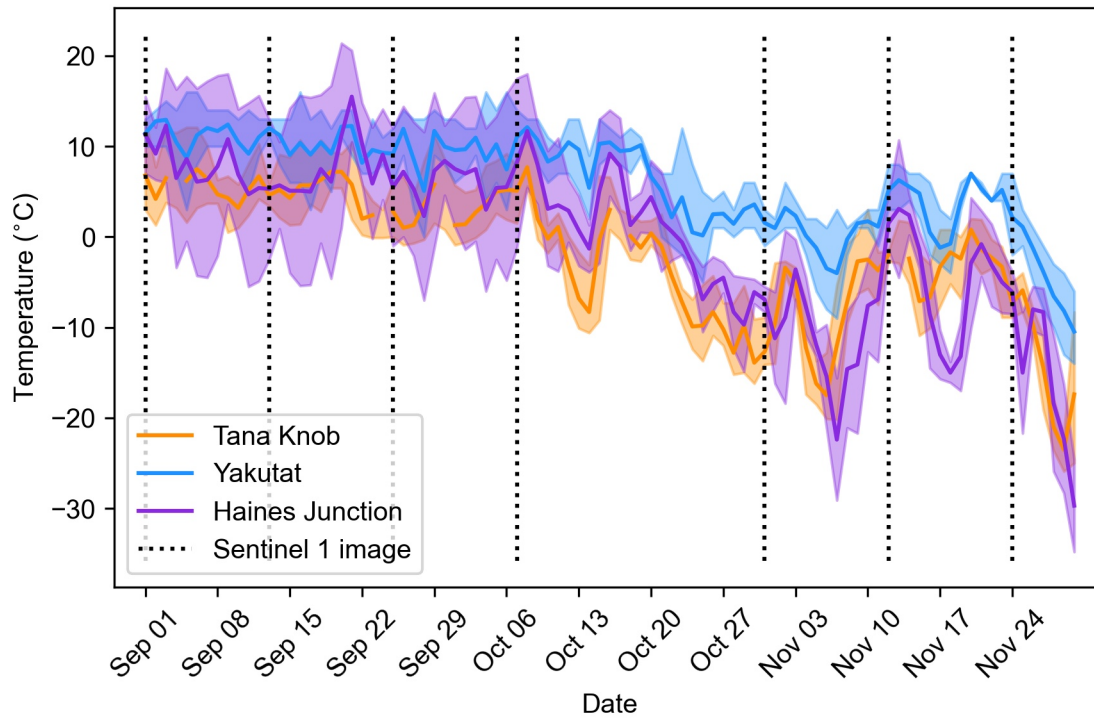


Fig. S20. Daily average temperature for Tana Knob, Yakutat, and Haines Junction for September 1st through November 30th, 2022. Shading ranges between the daily minimum and maximum. The vertical dotted lines mark days when Sentinel 1 imagery was collected for the selected path and frame. There was no mid-October, 2022 image for the selected path and frame.

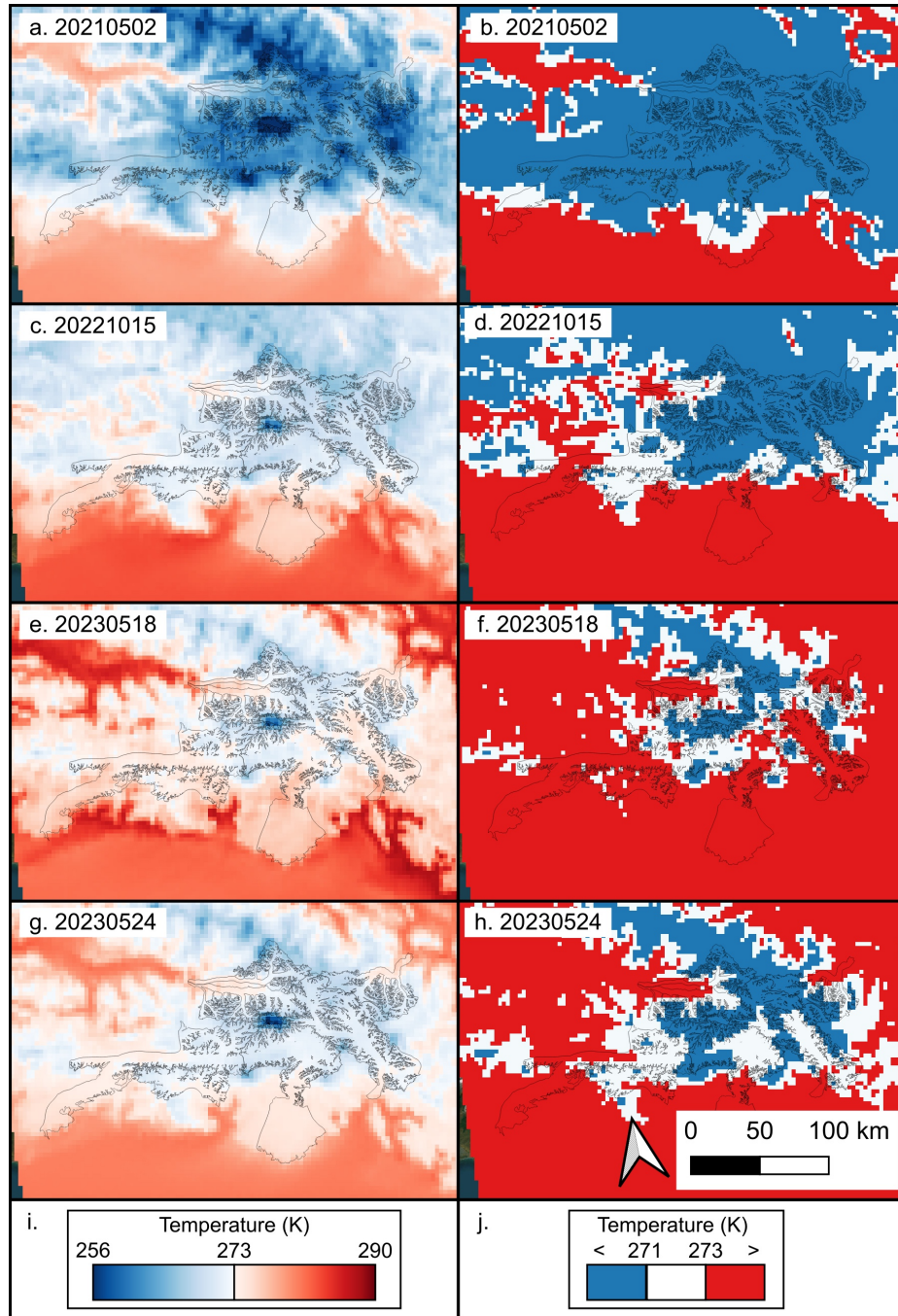


Fig. S21. CARRA2 daily average 2-m air temperature shows significantly warmer 2-m air temperatures during the 2018 and 2023 surveys than the 2021 survey. It also shows near freezing temperatures in parts of the Seward accumulation area on October 15, 2022. (a) CARRA2 daily average 2-m air temperature and (b) binned temperatures for May 5, 2021. (b) The binned temperatures are color coded such that colder than 271 K is blue, between 271 and 273 K is white, and warmer than 273 K is red. Similar plots for (c, d) October 15, 2022, (e, f) May 18, 2023, and (g, h) May 24, 2023. (i) Color bar for panels (a, c, e, g) show the temperature per grid cell for a temperature range of 256 to 290 K. Values below 256 K are shown as darkest blue. (j) Color bar for panels (b, d, f, h). Black outlines in panels (a)-(h) are the glacier outlines of the seven glaciers in this study.

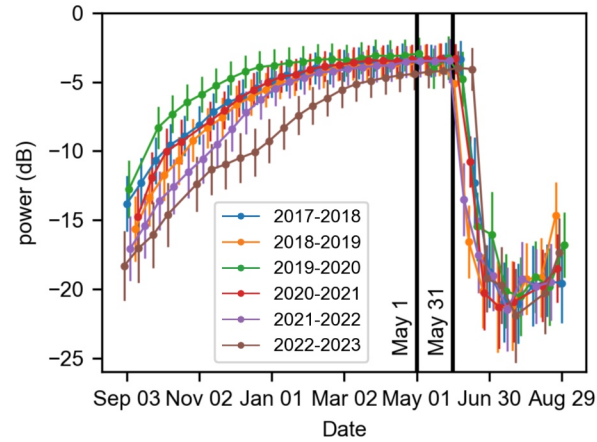


Fig. S22. Time series of Sentinel 1 reflectivity at the Kaskawulsh core site. Error bars are the one standard deviation of the power in the sampling area. Red line is the 2020-2021 accumulation season. No drop in reflectivity during fall 2020 is observed, which supports the interpretation of a strong, end of season ice layer that is not overlain by a fall rain on snow event. Also, the reflectivity drop from the onset of summer melting occurs later in the year than at the Seward sites due to the Kaskawulsh core site being at ~ 2650 m a.s.l., a kilometer higher than the Seward core sites.

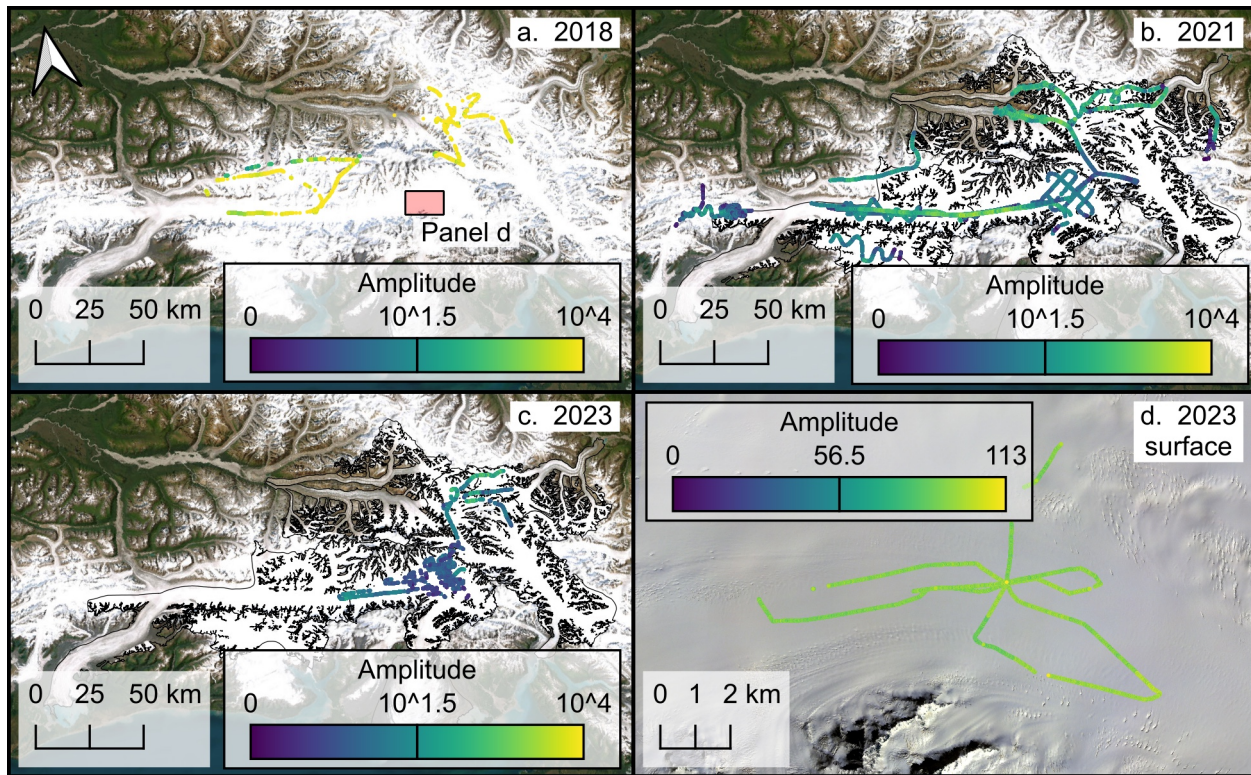


Fig. S23. Radar amplitude of the last summer surface for (a-c) the airborne radar and the (d) surface GPR. The scale for (a-c) the airborne radar is a logarithmic scale while the scale for the (d) surface GPR is a linear scale.

20 3 SUPPLEMENTARY TABLES

Table S1. Summary of snow pit and core collection date, pit depth, and core depth. Core depths are inclusive of the pit depths. Visualization of snow pit locations in Fig. 2.

Collection date	Latitude (°N)	Longitude (°W)	Pit	Core	Pit Max Depth (m)	Core Max Depth (m)
May 15, 2023	60.381139 N	140.277101 W	1	1	2.4	7.41
May 15, 2023	60.381139 N	140.277101 W	1	2	2.4	5.84
May 15, 2023	60.387577 N	140.292982 W	2	1	1.6	6.21
May 17, 2023	60.375174 N	140.245856 W	3	1	1.7	6.95

Table S2. Accumulation area mass input in gigatons (Gt) of water for each glacier using the radar only estimation and the sensitivity test including the interpolation to the Mt. Logan cores. Values are plotted in Fig. 8a. Uncertainties are reported as 2σ .

Glacier	Year	Gt (or 10^{12} kg) water; Radar only estimation	Gt (or 10^{12} kg) water; Mt. Logan core inclusion
Hubbard	2023	5.11 ± 0.46	5.07 ± 0.43
Kaskawulsh	2023	1.47 ± 0.23	1.27 ± 0.17
Seward	2023	3.67 ± 0.34	3.64 ± 0.25
Bering	2021	2.69 ± 0.28	2.89 ± 0.21
Hubbard	2021	3.93 ± 0.24	3.89 ± 0.22
Kaskawulsh	2021	0.93 ± 0.15	0.93 ± 0.13
Logan	2021	0.60 ± 0.11	0.60 ± 0.05
Seward	2021	3.69 ± 0.33	3.49 ± 0.25
Walsh	2021	0.41 ± 0.06	0.44 ± 0.02
Yahtse	2021	2.55 ± 0.15	2.44 ± 0.11
Bering	2018	2.49 ± 0.17	2.48 ± 0.17
Hubbard	2018	5.21 ± 0.42	4.99 ± 0.38
Logan	2018	0.95 ± 0.05	0.88 ± 0.04
Walsh	2018	0.80 ± 0.18	0.54 ± 0.04

Table S3. Mean accumulation area mass input in gigatons (Gt) of water for each glacier using the radar only estimation and the sensitivity test including the interpolation to the Mt. Logan cores. Values are plotted in Fig. 8b. Uncertainties are reported as 2σ .

Glacier	Gt (or 10^{12} kg) water; Radar only estimation	Gt (or 10^{12} kg) water; Mt. Logan core inclusion
Bering	2.59 ± 0.16	2.69 ± 0.13
Hubbard	4.74 ± 0.22	4.65 ± 0.20
Kaskawulsh	1.20 ± 0.14	1.1 ± 0.11
Logan	0.78 ± 0.06	0.74 ± 0.03
Seward	3.68 ± 0.24	3.56 ± 0.18
Walsh	0.60 ± 0.09	0.49 ± 0.02
Yahtse	2.55 ± 0.15	2.44 ± 0.11

Table S4. For each glacier and survey year the percent of the accumulation area sampled by the airborne radar, and percent of the accumulation area that is above the lowest surface elevation with airborne radar data, the total area below, within, and above the airborne radar. Total areas reference the total glacier (accumulation and ablation) area. Visual representation of this data is in Fig. S10.

Glacier	Year	Percent of							
		Percent of accumulation area sampled by radar	Percent of accumulation area above the lowest elevation with radar	Total area below radar (km ²)	Total area below radar (%)	Total area sampled by radar (km ²)	Total area sampled by radar (%)	Total area above radar (km ²)	Total area above radar (%)
Hubbard	2023	58.37	63.28	1119.62	39.5	1581.65	55.8	133.22	4.7
Kaskawulsh	2023	61.86	89.74	463.64	44	406.74	38.6	183.35	17.4
Seward	2023	68.63	92.02	1735.13	51.6	1213.92	36.1	413.61	12.3
Bering	2021	57.64	100	1107.19	36.6	1213.07	40.1	704.85	23.3
Hubbard	2021	72.28	78.14	717.13	25.3	1958.63	69.1	158.73	5.6
Kaskawulsh	2021	78.69	100	210.74	20	702.83	66.7	140.14	13.3
Logan	2021	36.48	100	204.84	17.4	513.28	43.6	459.13	39
Seward	2021	72.62	100	1593.9	47.4	1284.54	38.2	484.22	14.4
Walsh	2021	17.07	100	170.98	23.8	177.45	24.7	369.98	51.5
Yahtse	2021	53.64	92.51	138.6	13.6	510.57	50.1	369.93	36.3
Bering	2018	94.55	96.1	1424.83	47.1	1573.06	52	27.23	0.9
Hubbard	2018	57.43	63.28	1119.62	39.5	1556.14	54.9	158.73	5.6
Logan	2018	75.73	100	323.74	27.5	678.09	57.6	175.41	14.9
Walsh	2018	11.27	94.2	298.14	41.5	50.29	7	369.98	51.5

Table S5. Hyperbolic analysis results for the 1 m deep probe in Snow Pit 1 on the Seward Glacier. Data were collected on May 20, 2023.

Track	Time window (ns)	TWTT (ns)	Relative permittivity
44	500	12.5	3.51
45	100	12.744	3.65
46	100	12.363	3.43
47	100	12.461	3.49
Mean	—	12.517	3.52
TWTT stdev	—	0.162	—

Table S6. Hyperbolic analysis results for the 1.5 m deep probe in Snow Pit 1 on the Seward Glacier. Data were collected on May 20, 2023.

Track	Time window (ns)	TWTT (ns)	Relative permittivity
48	500	16.653	2.74
49	100	16.65	2.77
Mean	–	16.607	2.75
TWTT stdev	–	0.062	–

Table S7. Hyperbolic analysis results for the 1.2 m deep probe in Snow Pit 3 on the Seward Glacier. Data were collected on May 17, 2023.

Track	Time window (ns)	TWTT (ns)	Relative permittivity
32	500	13.477	2.83
33	500	13.232	2.73
Mean	–	13.355	2.78
TWTT stdev	–	0.17	–

Table S8. Data for the mean, median, standard deviation, in addition to the accumulation area ratio and total glacier area. The mean snowline is used as the ELA.

Glacier	Mean snowline (m)	Median snowline (m)	stdev snowline (m)	Accumulation area ratio	Total glacier area (km ²)
Bering	1421	1429	41	0.55	3025
Hubbard	1061	1064	24	0.96	2834
Kaskawulsh	2106	2140	100	0.62	1054
Logan	2112	2127	45	0.61	1177
Seward	1201	1209	102	0.53	3363
Walsh	2234	2247	57	0.62	718
Yahtse	912	874	91	0.93	1019

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4 UNCERTAINTY PROPAGATION

4.1 Converting from two way travel time to other variables

Equations 1–4 were used to convert two way travel times to real permittivity of layer depth. Real permittivity (Eq. 1) and associated uncertainty (Eq. 2) when two way travel time and layer depth are known. Layer depth (Eq. 3) and associated uncertainty (Eq. 4) when two way travel time and real permittivity are known.

$$\epsilon' = \left(\frac{t \cdot 0.3 \text{ m ns}^{-1}}{2 \cdot z} \right)^2 \quad (1)$$

$$\delta\epsilon = 2 \left(\frac{0.3 \text{ m ns}^{-1}}{2 \cdot z} \right)^2 \cdot t \cdot \delta t \quad (2)$$

$$z = \frac{t \cdot 0.3 \text{ m ns}^{-1}}{2 \cdot \sqrt{\epsilon'}} \quad (3)$$

$$\delta z = \sqrt{\left(\frac{\delta t \cdot (0.3 \text{ m ns}^{-1})}{2 \cdot \sqrt{\epsilon'}} \right)^2 + \left(\frac{(-1) \cdot \delta\epsilon' \cdot t \cdot (0.3 \text{ m ns}^{-1})}{4 \cdot \epsilon'^{(3/2)}} \right)^2} \quad (4)$$

Variables:

ϵ' : real permittivity

$\delta\epsilon'$: real permittivity uncertainty

t : two way travel time (ns)

δt : two way travel time uncertainty (ns)

z : layer depth (m)

δz : layer depth uncertainty (m)

4.2 Fifty meter elevation bin and input mass uncertainty

The following steps were used for uncertainty propagation when calculating the SWE per elevation bin and the accumulation area mass input.

1. When calculating the mean two way travel time for each 50 m elevation bin, we also calculated the

standard deviation. We used two standard deviations as the uncertainty in this propagation.

2. Two standard deviations from the mean crossover analysis were added to the uncertainty from step one.

3. Snow density for each segment uses a 9% uncertainty, which is propagated to the column average uncertainty. Eq. 5 is used for the uncertainty on each Seward core, while Eq. 6 is used to calculate the mean between the Seward core mean and the Kaskawulsh core density. The resulting column average uncertainty from both site is 22 kg m^{-3} . We use two times this average for the uncertainty on snow density.

$$\bar{dp} = \frac{1}{N} \sqrt{\sum_N dp_k^2} \quad (5)$$

$$\bar{d\rho} = 0.5 \cdot \sqrt{dx^2 + dy^2} \quad (6)$$

Variables:

$\bar{d\rho}$: final snow density uncertainty; 22 kg m^{-3}

dx : Seward core uncertainty; 38 kg m^{-3}

dy : Kaskawulsh core uncertainty; 21 kg m^{-3}

4. The average snow depth and SWE per elevation bin was then calculated. The average snow depth and snow depth uncertainty per elevation bin was calculated using Eqs. 3 & 4. The snow depth per bin was converted to SWE per bin using Eqs. 7 & 8.

For the real permittivity uncertainty we used two times the standard deviation from the nine equations in Sugiyama and others (2010) for a density of 438 kg m^{-3} which provided $1.86 \pm 0.16 (2\sigma)$. As there is the uncertainty in the snow density ($44 \text{ kg m}^{-3} (2\sigma)$), we also calculated the real permittivity for $394 \text{ kg m}^{-3} (1.77)$ and $482 \text{ kg m}^{-3} (1.95)$, which is a ± 0.09 spread from the real permittivity calculated using the average snow density. We added this value to the uncertainty for the average density providing a resulting in a real permittivity and uncertainty of 1.86 ± 0.25 to be used in the bin calculations.

For the two way travel time uncertainty, we used two times the standard deviation of all two way travel time measurements in each 50 m elevation bin. We added to that the mean crossover analysis result of

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62 1.49 ns.

$$s = z \cdot \rho \quad (7)$$

$$\delta s = \sqrt{\left(\frac{\delta z}{z}\right)^2 + \left(\frac{\delta \rho}{\rho}\right)^2} \cdot s \quad (8)$$

63 Variables:

64 s: average SWE per elevation bin (m)

65 δs : average SWE per elevation bin uncertainty (m)

66 z: average snow depth per elevation bin (m)

67 δz : average snow depth per elevation bin uncertainty (m)

68 ρ : snow density to water density ratio (0.438)

69 $\delta \rho$: snow density uncertainty (0.044). 0.044 from the snow density uncertainty divided by the density
70 of water ($44 \text{ kgm}^{-3}/1000 \text{ kgm}^{-3}$).

71 5. The mass uncertainty per elevation bin is calculated from converting the average SWE per elevation bin
72 to units of mass.

$$\delta m = \delta s \times A \times 10^{-3} \quad (9)$$

73 Variables:

74 δm : mass uncertainty per elevation bin (m)

75 δs : SWE uncertainty per elevation bin (m)

76 A: Glacier area from GLIMS (RGI Consortium, 2023) (km)

77 10^{-3} factor: needed for converting m SWE to gigatons. 1 Gt = 10^{12} kg. Converting m to kg uses a
78 factor of 10^9 .

79 6. Then, the uncertainty per elevation bin is converted to the total accumulation area mass input uncer-
80 tainty though Eq. (10).

$$\overline{M} = \sqrt{\sum_N \delta m^2} \quad (10)$$

:

Variables:

$\overline{M}$: accumulation area mass input uncertainty (m)

δs : mass uncertainty per elevation bin (m)

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OpenTopography (2021) Copernicus GLO-90 Digital Surface Model (doi: 10.5069/G9028PQB)

RGI Consortium (2023) Randolph Glacier Inventory - A Dataset of Global Glacier Outlines, Version 7 (doi: 10.5067/F6JMOVY5NAVZ)

Sugiyama S, Enomoto H, Fujita S, Fukui K, Nakazawa F and Holmlund P (2010) Dielectric permittivity of snow measured along the route traversed in the Japanese–Swedish Antarctic Expedition 2007/08. *Annals of Glaciology*, **51**(55), 9–15, ISSN 0260-3055, 1727-5644 (doi: 10.3189/172756410791392745)